Khojki OCR | Technical retrospective

Round four - source audit, controlled experiments and frozen evaluation - 31 October 2026

NEW FROZEN TEST CER	ADDITIONAL TRAINING RUNS	CPU PAGE TIME
17.50%	**9**	**3.28 s**

Pipeline and honest outcome

Page/PDF -> background-relative ink and frame removal -> lines and number-column chunks -> two 1.06M CNN-BiLSTM-CTC recognizers -> structural beam decoding -> UTF-8 and offline correction UI. HarfBuzz/FreeType shape Noto synthetic data; selected weights adapt on 15 real printed lines. Inference needs no API or retraining.

New frozen test: 21/120 edits = 17.50% CER on five real page-21 lines. The previous model has the same result: nine additional training runs did not beat it on development, so it was retained. Development remains 14/405 (3.46%), an optimistic selection score. All five source/chart references were frozen before predictions; two uncertain readings were declared in advance (both alternatives leave 17.50%). Related print and non-expert references limit generalization claims.

Errors by stage

0 missed selected lines; 0 space-involving edits; 21 other recognition edits; 0 invalid code points / 0 UTF-8 failures. Automatic and checked crops yield identical predictions. Valid-but-wrong letters evade encoding validation. These are operational categories, not a uniquely identified causal allocation; unlabelled regions are unscored. Three alignments pair inserted PA with E-to-I substitution; E has no real training example.

What failed, with numbers

Initial synthetic CER was 1.31%, but real-line CER was 48%; contrast/chunking reduced it to 29.33%. This round, six matched 600-step pilots gave 16-17/405 edits versus incumbent 14. A confirmed RGB-padding bug created red/dark borders: both padding variants gave 16 edits at 600 steps; at 3000, old artifact 14 versus fixed white 16. Correct image handling did not imply an accuracy gain. Slower 1000-step original/corrected-label/10%-word runs gave 16/18/18; none qualified for 5000. Five contrast/blur variants gave 15-18. Full histories and failed runs are retained.

Encoding decisions

Dependent vowels: U+1122C-U+11233; anusvara U+11234; independent AA U+11201. O/AU signs are atomic U+11232/U+11233. Word separator: U+1123A; printed digits: Gujarati U+0AE6-U+0AEF. ASCII space marks the distant number column. CER counts Unicode scalars, including signs and digits. Inventory, orphan/adjacent signs, atomic vowels, NFC changes and UTF-8 round trips are checked; raw text is not silently repaired.

AI error caught in our own labels

Image-only repeated-word matching plus enlarged scan/chart comparisons exposed likely errors in earlier AI-authored references (for example, U versus II and BA versus NA). Four proposed word revisions change the same old output from **26/123 to 15/128 edits**. This is a post-hoc reference effect, **not better OCR**; originals remain frozen and expert confirmation is needed. One training-only sign correction was tested separately and did not improve development. Earlier VLM probes also returned Kannada/Devanagari (2/2 wrong-script failures).

Single biggest bottleneck

Obtain **200-500 independently adjudicated printed lines** spanning books, fonts, damage and rare signs; double-annotate a subset and measure a 50/100/200/500-line curve with book-disjoint tests. This is a planning estimate. Thirty-four supported Khojki characters have no examples in the 15 real training lines. Modern-font rendering cannot adjudicate every historical form. Initial 31-line authoring/freezing took about 14 minutes plus chart study and review; this is AI-assisted elapsed time, not expert productivity.

Scale, resources and handwriting

3.28 s per 16-line page on a two-thread Colab CPU, plus 0.045 s loading; CPU/CUDA text matched. Roughly 154 s for 47 comparable printed pages, excluding startup/PDF rendering and review. No equivalent quality is claimed for the mixed collection. T4 was about 1.15 units/hour; setup/idle time dominated. All archives verified; 166.94 units remain, zero sessions. Handwriting needs baseline/polygon detection, writer-disjoint evaluation and perhaps 1,000-3,000 diverse labelled lines initially; current handwriting output is unvalidated.